PAPER_693: Sombrero Galaxy (M104/NGC 4594): Edge-On Sa Dynamics and UQFF

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Abstract

The Sombrero Galaxy (M104/NGC 4594) is a nearby (8.6 Mpc) nearly edge-on Sa galaxy with an exceptionally massive bulge, prominent dust ring, and a $10^9 M_\odot$ central black hole. Its dynamics are modeled via UQFF-modified gravitational potentials.

1. System Parameters

Parameter	Value
M_{bulge}	$8 \times 10^{11} M_\odot$
M_{disk}	$2 \times 10^{11} M_\odot$
M_{BH}	$10^9 M_\odot$
R_{eff}	4.2 kpc
Inclination	84°

2. Circular Velocity

$$v_c(R) = \sqrt{\frac{GM_{enc}(R)}{R}} \left(1 + \frac{\rho_{SCm}}{2\rho_{UA}}\right)$$

3. Gravitational Potential (UQFF-modified)

$$\Phi_{UQFF}(r) = -\frac{GM_{bulge}}{r+a} - \frac{GM_{disk}}{2R_d} \ln(R^2 + h^2) + \frac{\Lambda c^2 r^2}{6}$$

where $a = R_{eff}$ is the Hernquist scale radius.

4. Bulge Velocity Dispersion

$$\sigma_{bulge} = \sqrt{\frac{GM_{bulge}}{5R_{eff}}} \approx 230 \text{ km/s}$$

5. Dust Lane Wavefunction

$$\Psi_{dust}(z,t) = e^{-z^2/(2h^2)} \cos(\omega_i t)$$

References

- Kormendy et al. (1996), ApJ, 473, L91
- UQFF Framework, Star-Magic Session 174

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